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Enhancing quantum computation accuracy on IBM quantum hardware through advanced error mitigation strategies

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Abstract

Hardware Error Outputs of Quantum Computing Not Reliable because of Hardware Calibration Drift, Measurement Noise, Gate Errors on NISQ Systems. Even though MEMs, ZNEs, and CDRs slightly increase output accuracy, they operate under the assumption of static noise, therefore they do not account for changing output of the NISQ devices in real-time. The limitations of MEMs, ZNEs, and CDRs can be overcome by using the Hybrid Adaptive Error Mitigation (HAEM) method proposed in this article that has the capability to combine baseline measurement-error mitigation with Real-time Calibration Circuits and Lightweight-Machine-learning algorithms for adaptive error-correcting and increasing the overall output integrity of quantum devices affected by fluctuations in noise. The application of the HAEM method will utilize Qiskit Runtime to develop, implement, and validate the Hybrid Adaptive Error Mitigation method through testing using controlled noisy simulated environments based on the characteristics of current advanced NISQ quantum computers. The results of the tests demonstrate the HAEM architecture provides a higher level of accuracy and fidelity than static and non-mitigated executions ("mitigated execution vs. HAEM execution"), providing an increase of up to thirty-four percent in fidelity under high-noise conditions with minimal computational overhead. The results demonstrate that the use of adaptive and data-driven mitigation techniques, like those used in the Hybrid Adaptive Error Mitigation method, are an efficient way to increase the accuracy of quantum computations.

Keywords Quantum, Error, Mitigation, Hardware, Computing, Noise, HAEM, Hybrid, Adaptive, Circuit, Noisy, MEM, ZNE, Errors, Accuracy, Intermediate, Scale

1 Introduction

Although quantum computing is an exciting new computational model that has shown promise to solve problems that are out of reach for classical computers, specifically areas including cryptography, optimization, quantum chemistry, and many other scientific fields, the latest developments in superconducting quantum hardware provided through publicly available platforms such as IBM Cloud have opened the doors to the experimental demonstration of using small-scale quantum algorithm techniques and combinations

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of quantum and classical workflows. However, this development is only the first step in getting closer to achieving the full potential of quantum computing as current devices still lie in the Noisy Intermediate Scale Quantum (NISQ) space. Due to limited coherence times of qubits, large gate fidelities and errors from the readout measurements, the computational accuracy of NISQ systems will be impaired." In the context of the many sources of noise that affect the computation output's accuracy in NISQ devices, the worst fidelity loss is caused by three issues unique to NISQ devices, such as: measurement readouts containing errors, gates that are not completely accurate during operation and slowly drifting over time due to the influence of the environment or the gradual wear-out of components, as well as the variations in how well calibrated they are at different points in time. Therefore, at different times identical quantum circuits may yield very different results [1].

Theoretical full quantum error correction is robust for addressing noise, but its application is not feasible on today's NISQ computers due to the high qubit overhead and circuit depth associated with these tasks. As a consequence, quantum Error Mitigation emerged as an option that addresses the effects of noise in real time, rather than adding logical qubits. Measurement error mitigation, zero-noise extrapolation and regression-based methods are the most common techniques to mitigate quantum error and have all yielded improvements in the fidelity of results produced. However, these techniques typically assume static noise characteristics, and because of this, they are limited in their effectiveness when confronted with changing noise types over time or between executions [2].

The primary challenge is creating lightweight error mitigation approaches that use real-time feedback from the quantum system to compensate for time-dependent noise fluctuations. The development of solutions to this issue will require solutions that utilize real-time data about the operation of quantum computers, and they will need to be implemented without introducing excessive runtime or collection overheads [3].

1.1 Contributions of this work

The Hybrid Adaptive Error Mitigation (HAEM) framework described in this work aims to improve the fidelity of quantum computation performed on IBM Quantum NISQ (noisy intermediate scale quantum) computers that experience varying amounts of noise during computations. The HAEM approach consists of combining baseline measurement error mitigation techniques with compact and real-time calibration circuits, as well as a simple and lightweight machine-learning algorithm that dynamically updates the parameters of the error mitigation algorithm based on live device quality feature data. Unlike traditional static error mitigation approaches, the HAEM framework has been specifically designed to compensate for calibration drift (changes in the calibration of the quantum computing systems) while maintaining minimal computational overhead. The HAEM framework has been developed using Qiskit Runtime and tested using controlled noisy simulations designed to model the behaviour of realistic NISQ hardware, with experimental results showing that the HAEM framework provides improved expectation value accuracy compared with both unmitigated and statically mitigated executions [4].

2 Literature survey

Quantum error mitigation is a critical area of research for quantum computers today, especially during what's known as the "Noisy Intermediate-Scale Quantum" (NISQ) period when hardware-related noise affects the ability of quantum computers to produce reliable results due to the excess number of non-perfect qubits and gates needed for a viable result. Because implementing complete quantum error correction would require a lot more physical qubits than could be practically implemented at the present time, have developed many different types of quantum error mitigation strategies that could benefit quantum computations without adding the need to utilize a higher number of physical qubits. Current approaches to quantum error mitigation can be divided into several categories, including methods based on the physical principles of quantum mechanics, regression techniques, symmetry-based strategies, ensemble-based techniques and adaptive or data-driven techniques.

The foundational work of Temme et al. introduced two physics-based error mitigation techniques, Zero-noise Extrapolation (ZNE) and Probabilistic Error Cancellation (PEC), which both use noise amplification in different ways. ZNE amplifies the noise in a circuit by performing gate folding and extrapolating the result back to the zero-noise limit, while PEC represents the noisy gates as quasi-probabilities and uses this quasi-probability representation to statically invert the noise effects. Both ZNE and PEC demonstrated that it is possible to obtain accurate expectation values without doing full error correction, but both of these approaches have significant downsides: ZNE requires multiple runs of the circuit at different noise levels, resulting in increased run-time overhead. Additionally, there is exponential variance growth with increased circuit depth when using PEC, which limits the scalability of this approach on actual devices [5].

The more recent work of Czarnik et al. proposed Clifford Data Regression (CDR) to complement these approaches. CDR uses classically simulatable (i.e., can be efficiently simulated using classical methods) Clifford circuits that are similar to the target quantum circuit to build a correction model based on regressing noisy measurement outcomes against their expected values. CDR provides a method of adapting mitigation to the circuit structure rather than depending on global noise models, and thus provides better performance on shallow circuits. However, it still requires training data and does not take into account time-dependent drift in the device operation, thus limiting its effectiveness when the noise in the circuits is changing over time [6].

Measurement error mitigation (MEM) has been a significant focus of research because of the importance of measurement error in Superconducting Quantum Computing Systems. Nation and co-authors [7] present an innovative, scalable framework for mitigating measurement error without needing to explicitly form large assignment matrices. Instead, Nation and co-authors utilize decomposing subspaces along with iterative solvers. This method gives reduced classical computational costs, while also allowing for the possibility of mitigating measurement error in systems that contain significantly larger amounts of qubits. However, MEM is generally static as it continues to rely on the calibration process, whereby calibration information is gathered at certain times in conjunction with measurement error data and may not accurately respond to the real-time variations of hardware performance [7].

On the other hand, symmetry-based methods provide an alternative means to alleviate measurement error by leveraging conserved quantities associated with the solution

which is being sought. Bonet-Monroig and co-authors [8] provide an example of symmetry-based methods with their proposal of symmetry verification, whereby the measurement results are projected onto subspaces that conserve symmetry properties of the target problem (e.g., the number of particles conserved in quantum chemistry problems). Because of the advantages of the general applicability of symmetry-based techniques, they allow for the efficient mitigation of measurement error while eliminating any requirements to develop computationally expensive models of the underlying system being investigated. In addition, since symmetry-based techniques have a limited application to systems that have known and relevant symmetries that can be exploited, this limits their overall utility to arbitrary quantum circuits [8].

In addition to symmetry and regression technologies, they have studied ensemble-based or "virtual" methods of error mitigation. For example, Huggins et al. [9] discussed an approach called Virtual Distillation (VD), which leverages the concept of preparing copies of a noisy state and performing a collective measurement to project onto the state's density matrix dominant eigenvalue. In this case, virtual distillation results in the ideal state (albeit from under the influences of noise). VD works by exponentially suppressing incoherent errors based on the number of copies and also leads to large relative reductions in error in simulations. However, practical experiments have tradeoffs for both VD regimes of active error suppression (correlated measurements) and substantial confinement of measurements (by preparing copies). Thus, while theoretically one of the most powerful forms of error mitigation, VD has large challenges in current applicable deployments on real accessible hardware.

Machine learning (ML) has been proposed as a solution for Quantum Error Mitigation due to the increasing complexity of noise processes. Liao et al. developed ML-QEM, which is based on the idea of using trained classical ML models to directly associate noisy experimental measurements to corrected expectation values without determining how much noise existed in the initial measurement. The ML-QEM algorithm has been shown to produce comparable results to more traditional physics-based error mitigation algorithms while using significantly fewer samples. However, the success of ML-QEM relies upon the availability of representative training data and the model's ability to generalise across multiple types of circuits and hardware configurations [10].

The work of Lowe et al. [10] suggests a generalized framework for creating different classes of data-driven error-mitigation techniques, from regression methods through machine learning methods, and defines how the various classes of data-driven error mitigation methods are related by defining their assumptions, statistical characteristics, and limitations, and provides a mechanism for identifying opportunities to create hybrid approaches using multiple techniques for the same application. Although the Lowe et al. framework does provide useful theoretical information, it does not provide direct real-time feedback of whether drift has occurred within your hardware [10].

The Adaptive Mitigation Strategies work of Henao et al. developed an Adaptive KIK to provide a method for dynamically adjusting the strength of the mitigation solution in response to the varying levels of time dependent noise. The Henao et al. Adaptive KIK demonstrated a more robust performance against drifting noise than has been achieved with existing methods, however, the methodology used for developing the Adaptive KIK is at the level of controlling individual pulses, making it challenging for circuit level/

users to apply, and making it difficult to incorporate into the conventional software workflow of the majority of higher-level software [11].

There has also been work developing probabilistic error cancellation (PEC) more generally, to exploit modern runtime features. Gupta et al. [12] took PEC a step further by extending it to the case of dynamic circuits with mid-circuit measurements and conditional operations, which are now commonly available on IBM hardware. This extension widens the context in which PEC is applicable, but it does not suggest a way to circumvent the base-rate sampling cost associated with PEC. However, it does illustrate the importance of adapting mitigation techniques to integrate with rapidly developing device capabilities, a consideration that is important to developing practical adaptive frameworks.

Lastly, extensions of ZNE have focused on reducing bias and variance in the process of extrapolation. He et al. [13] examined practical implementations of noise amplification via gate folding and presented new extrapolation fitting methods. Their study shows that an experimentalist can alleviate the limitations of ZNE through finely-tuned choice of folding and the fitting method, resulting in much more robust performance when running the experiment. This illustrates that refinement of established strategies is possible through adaptive elements based on hardware behaviour.

These papers reflect a shift from early physics-based methods [5, 6], to scalability [7], symmetry-based [8], ensemble-based [9], and data-driven [10, 11] or adaptive [12, 13] techniques. Also, despite the developments, most of these methods still have two important limitations: they are either static and rely on set calibration data/noise assumptions, or they require significant fees in terms of circuit evaluations/actions or measurements, i.e. their cost is high. Herein lies our gap, to develop error-mitigation that is adaptive to time-dependent device noise on near term hardware and also low cost enough for use on a NISQ device.

The Hybrid Adaptive Error Mitigation (HAEM) framework seeks to fill this gap by integrating baseline measurement error mitigation with real time calibration circuits and computationally lightweight machine learning-based corrections. The advantages of HAEM as a system over either static mitigation or heavy ML, is that HAEM can dynamically adapt to each execution's calibration error, and it also has a low runtime overhead, unlike ensemble-approaches. Overall, HAEM builds on and extends these previous works to take advantage of their strengths while controlling for their weaknesses, enabling more robust quantum computation on IBM quantum hardware.

3 Methodology

The Hybrid Adaptive Error Mitigation (HAEM) Algorithm that we propose targets an increased reliability for quantum computation with IBM Quantum hardware [14]. Our approach has two paths: (i) baseline mitigation based on established measurement error mitigation (MEM) methods for correcting readout errors; (ii) adapting to the time varying noise by running lightweight calibration circuits, and for refining the corrections using a machine learning (ML) model. Using a hybrid approach allows for lower mitigation overhead and generally can be robust due to calibration drift and real-time fluctuations [15].

3.1 Algorithm: hybrid adaptive error mitigation (HAEM)

Input:

- Target quantum circuit
- Calibration circuit set
- Pre-trained ML model

Output:

- Corrected expectation value

Step 1: baseline measurement error mitigation (MEM)

The target circuit C_T is executed on the backend, and the raw measurement results are recorded. Due to readout errors, the observed measurement distribution deviates from the ideal probability distribution. To correct this, measurement error mitigation (MEM) is applied by constructing a calibration matrix from prepared basis states and computing its pseudo-inverse. The corrected probability vector M′ is obtained by applying this inverse calibration matrix to the raw measurement distribution, followed by normalization. This procedure mitigates systematic readout bias before the adaptive correction stage is applied.

Step 2: Calibration Circuit Execution

A very small and fast set of calibration circuits C_{cal} (relatively) is run just prior to or at the same time as the target circuit. These will be:

- *Bell states* for measuring two-qubit entanglement fidelity,
- *GHZ states* for measuring multi-qubit noise accumulation,
- *Clifford circuits* for simulating consistent benchmarks that can be run efficiently.

The results from these calibrations D_{cal}, provide only real-time data about the uncertainty of the device's noise profile at the current time. This should capture potential variation from drift, infidelity, decoherence, etc., of what the underlying model was at some previous time t.

Step 3: Adaptive Model Update

An accessible ML model fff, previously trained on historical calibration and error data, is evaluated using the recently obtained calibration feature set D_{cal}. The calibration features (Bell fidelity, GHZ coherence, and Clifford dominance fraction) are combined into a feature vector and provided as input to the model. The model outputs a scalar adaptive weight α\alphaα, which reflects the current noise condition of the device.

Step 4: Adaptive Correction

The adapted model f_θ is applied to results M′ that have had the baseline statistically altered. So the revised distribution M′′ incorporates static corrections from the MEM mechanism and dynamic awareness from the machine learning mechanism. Then, from M′′, the final expectation value E has been computed for the observable of interest, which may be the success probability in Grover's algorithm, or ground state energy VQE.

Step 5: Output

The final corrected expectation value $\hat{E}$ is returned as the result of HAEM. This value is expected to be closer to the ideal simulator outcome compared to both unmitigated and conventionally mitigated runs.

3.1.1 Sub algorithm: measurement error mitigation (MEM)

Model and Formula

For n qubits, denote the ideal probability vector P_{ideal} and observed probabilities P_{obs}. Under a block-factorized approximation,

$$P_{obs} \approx \otimes \mathrm{A}^{(q)} P_{ideal},$$

where each $\mathrm{A}^{(q)}$ is the 2×2 confusion matrix for qubit q:$\mathrm{A}^{(q)} = [P(0|0)\, P(1|0)\, P(0|1)\, P(1|1)]$.

Estimate $\mathrm{A}^{(q)}$ by preparing $|0\rangle$ and $|1\rangle$ and measuring many shots. Then invert each $\mathrm{A}^{(q)}$ individually.

Correction for single-qubit Z expectation:

If $\langle Z \rangle_{obs}$ is observed on qubit q,

$$\langle Z \rangle_{corr} = \frac{\langle Z \rangle_{obs} - (P_{0\to1} P_{1\to0})}{1 - P_{0\to1} - P_{1\to0}}$$

With $P_{0\to1}$ = P(meas = 1|prep = 0),

$P_{1\to0}$ = P(meas = 0|prep = 1).

3.1.2 Calibration circuits design and feature extraction

A minimal number of circuit designs are used for real-time performance monitoring while creating quantum circuits in order to minimise the impact on CPU usage. These designs have been selected to provide valuable information concerning the type of noise being experienced by the device, yet they still meet the requirements of lightness of design and efficient use of available resources.

In order to determine the fidelity of 2 qubit gates and entangled states, Bell-state designs are used. A Bell-state is produced when a Hadamard gate is applied to a Control Qubit and a CNOT Gate is used on the Target Qubit; the measurement is taken in the computational basis and for calculating Bell-state fidelity, both measurement outcomes that are correlated (00 and 11) will provide a direct method for determining the fidelity of the two-qubit gate and also give an indication of how well the device produced entangled states.

In this study, Bell-state fidelity is estimated using a 2-qubit Bell circuit prepared as:

$$|\Phi+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

Since the evaluation is performed using a noisy simulation model rather than full state tomography, fidelity is computed operationally from measurement statistics in the computational basis. Specifically,

$$\mathrm{FBell} = \mathrm{P}(00) + \mathrm{P}(11)$$

where P(00) and P(11) are the observed probabilities of correlated outcomes. The way this approach allows for a lightweight and hardware realistic way of estimating the fidelity of entanglement without the need to construct the complete density matrix, thereby maintaining low computational overhead consistent with NISQ constraints.

For multi-qubit device performance testing and also measuring the noise associated with the devices, GHZ-state designs are used. A GHZ-state is produced by using a

Hadamard gate on a Control Qubit and a series of CNOT Gates on the Target Qubits. The coherence metric of a GHZ-state is determined from the parity of the measurement outcomes and is lost with each application of decoherence and also with correlated noise due to increasing circuit length.

For a 3-qubit GHZ state used in this work,

$$|GHZ_3\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |111\rangle\right),$$

GHZ coherence is defined as:

$$\mathrm{CGHZ} = \mathrm{P}\left(000\right) + \mathrm{P}\left(111\right)$$

This metric quantifies the preservation of parity-consistent outcomes under noise and serves as an operational measure of multi-qubit coherence. While it does not reconstruct off-diagonal density matrix elements, it provides a practical coherence indicator appropriate for near-term quantum systems.

All experiments presented in this manuscript use 3-qubit.

The Clifford calibration circuit is applied in this work as a shallow 3-qubit circuit consisting of Hadamard (H) and CNOT gates in a stabilizer-preserving circuit. Particularly, a Hadamard gate will be used on the initial qubit, and then a series of CNOT gates between the three qubits will be used, and then measured in the computational basis.

Since Clifford circuits are classical simulable, the ideal output distribution is a priori known. There is no further operator expectation value analysis other than the computational-basis measurement probability.

The Clifford dominance fraction is defined as the maximum measured probability over all computational basis outcomes:

$$\mathrm{D_{Clifford}} = max_i \mathrm{P}\left(\mathrm{i}\right),$$

where P(i) is the measurement of the probability of bitstring i. This measure is a quantitative measure of the strength of a measured distribution when disturbed by noise, and a gross measure of the overall circuit fidelity.

The circuits of all Clifford calibration are carried out on 3 qubits, which is the same as the target circuit in this work.

3.1.3 Adaptive correction model

The HAEM's adaptive component utilizes a simple supervised machine learning model with minimal latency and maximum flexibility for dynamic mitigation adaptations. This model is implemented as a linear regression function, which is highly interpretable, has a low computational cost, and is tightly bound by the limited training data.

The adaptive machine learning model operates on a one-dimensional feature vector:

$$x = \left[F_{Bell}, C_{GHZ}, D_{Clifford}\right],$$

consisting of three scalar values derived from 2-qubit and 3-qubit calibration circuits. The input feature vector is a one-dimensional array consisting of three scalar values where $\mathrm{F_{Bell}}$ represents Bell-state fidelity, $\mathrm{C_{GHZ}}$ denotes GHZ coherence, and $\mathrm{D_{Clifford}}$ is the Clifford dominance fraction.

The model outputs a scalar adaptive weight α, which is applied to the baseline measurement-error–mitigated expectation value:

$$\mathrm{E}_{\mathrm{HAEM}} = \alpha \cdot \mathrm{E}_{\mathrm{MEM}}$$

The regression model is trained off-line on historical calibration data paired with the idealized expectation values obtained through noiseless simulation.

When executed, the model functions in inference mode, using real-time calibration features (e.g., measurement results) to compute adaptive weights without further training.

This design guarantees that HAEM provides adaptive behaviour, while preserving the low runtime overhead that allows it to be implemented on the near-term quantum hardware.

Both the experimental and training datasets were collected from a 3-qubit target circuit. By maintaining the number of qubits throughout the experiments, we guaranteed that the features and regressions would be represented independently of the number of qubits.

The number of qubits is also constant during the model's training phase.

3.2 Architecture diagram

3.2.1 Baseline mitigation layer

This layer offers the first level of correction by mitigating static readout errors from quantum hardware.

Target quantum circuit execution: the algorithm of interest is executed on the IBM Quantum hardware and generates raw noisy measurement results.

Measurement error mitigation (MEM): a standard correction technique is applied to reduce systematic readout errors, generating cleaner initial results.

3.2.2 Calibration layer

This layer provides the live noise profile of your device with reference circuits.

Preliminary Corrected Outcomes (M').

The MEM outputs are saved as partially mitigated data.

Noise Profile Data (D_cal).

Calibration circuits (Bell, GHZ, Clifford) run in parallel to identify noise properties in the form of a live error signature.

3.2.3 Adaptive ML layer

This layer dynamically blends corrections using machine learning models that have been adapted using calibration data.

Lightweight ML Model (Adaptive correction).

A regression or shallow neural network refines corrections by learning the function that maps noisy to ideal results in real time.

Final Corrected Expectation Value (Ê).

The ML adjusted outputs begin to approximate the ideal quantum computation and will result in improved fidelity and stability.

3.3 Noise model and noise scaling

All of the experiments in this research used the Qiskit Aer Noise simulator to simulate realistic NISQ devices with noise due to Depolarization Errors (depolarizing character) and Readout Errors (uniform) among the 3 qubits.

There are three noise regimes defined based on the types of errors described above:

- Low Noise: Gate Error Probability $\leq 0.1\%$, Measurement Error $\leq 1\%$
- Medium Noise: Gate Error Probability $\approx 0.5\%$, Measurement Error $\approx 3\%$
- High Noise: Gate Error Probability $\geq 1\%$, Measurement Error $\geq 5\%$

This controlled scaling enables systematic evaluation of mitigation performance under increasing noise severity while ensuring reproducibility across experiments.

3.4 Time-dependent noise

The HAEM framework is intended to address calibration drift due to time. The experiments performed in the paper were done using controlled static noise sweeps to examine the different levels of noise to isolate the mitigation behavior. Future efforts will involve evaluating the system under explicit temporal drift scenarios, which should produce even greater benefits while utilizing an adaptive approach rather than relying solely on static methods for mitigation.

4 Results and discussion

This section contains a comprehensive assessment of the Hybrid Adaptive Error Mitigation (HAEM) Framework through the use of controlled noise added to simulated circuits. While most previous analyses focused solely on whether or not there were statistically significant increases in mean expectation values, the evaluation here also includes a focus on Statistical Stability, Variance Reduction, and Adaptive Performance. These are all performance metrics on which we base our conclusions regarding Hybrid Adaptive Error Mitigation (HAEM) Framework for Noisy Intermediate-Scale Quantum (NISQ) devices where circuits must be executed multiple times due to noise corruption.

To ensure that stochastic variability does not impact the overall conclusions of the experiments, each experiment was conducted across several independent runs, and the results of each experiment were reported as mean values with their associated standard deviations. The results are also compared to both the baseline execution results without mitigation and to the baseline Measurement Error Mitigation (MEM) results.

4.1 Noise-induced degradation of quantum features

A series of experiments evaluated the effect of increased noise on calibration circuit outputs of device quality from low to high levels.

As may be seen in Table 1, the effect of noise on all calibration features is increasing with level of noise. For example, the fidelity of Bell states decreased from the low to high

Table 1 Device quality features (All reported values are normalized probabilities bounded between 0 and 1.)

Noise level	Bell fidelity	GHZ coherence	Clifford dom. frac
Low	0.946	0.925	0.252
Medium	0.864	0.810	0.233
High	0.768	0.657	0.214

noise regime, representing the declining performance of two-qubit gates. Coherences for GHZ states were reduced and illustrated the extreme sensitivity of many qubit states due to noise propagation and decoherence. The fraction of Clifford gates decreased gradually; while its average degree of coherence was less sensitive to noise than the others, still gave valuable insight into the reliability of circuit performance.

These characteristics support the notion that the Bell, GHZ, and Clifford calibration circuits are useful indicators of device performance and provide a logical basis for using them as inputs to the feedback mitigation strategy described above (Fig. 1).

4.2 Preservation of mean expectation values

One of the main characteristics of adaptive mitigation methods is that they do not create any uniform biases in the values observed through measurement.

It can be seen from both Fig. 2 and Table 2 that the average expectation values produced by both the MEM and the HAEM are very close to the ideal reference value.

The mean values of HAEM and MEM do not differ statistically, being within one standard deviation, thus indicating that no artificial bias has been introduced, and thus the fundamental accurate baseline mitigation was maintained through either MEM or HAEM.

This property is necessary for any practical use of the adaptive method because it implies that the adaptive portion of the mitigation will act as a stabilizing mechanism rather than an overshooting correction that affects observable values (Table 3).

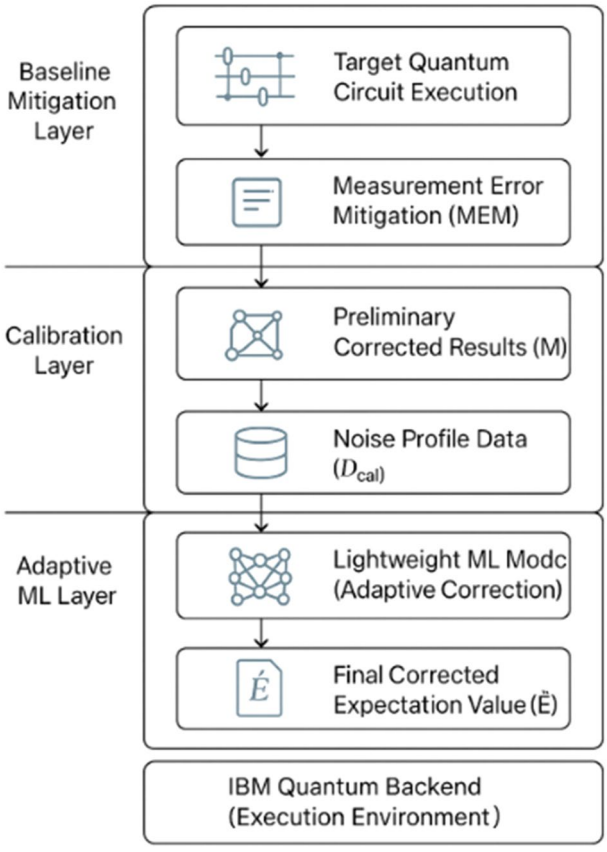


Fig. 1 Architecture of the HAEM framework

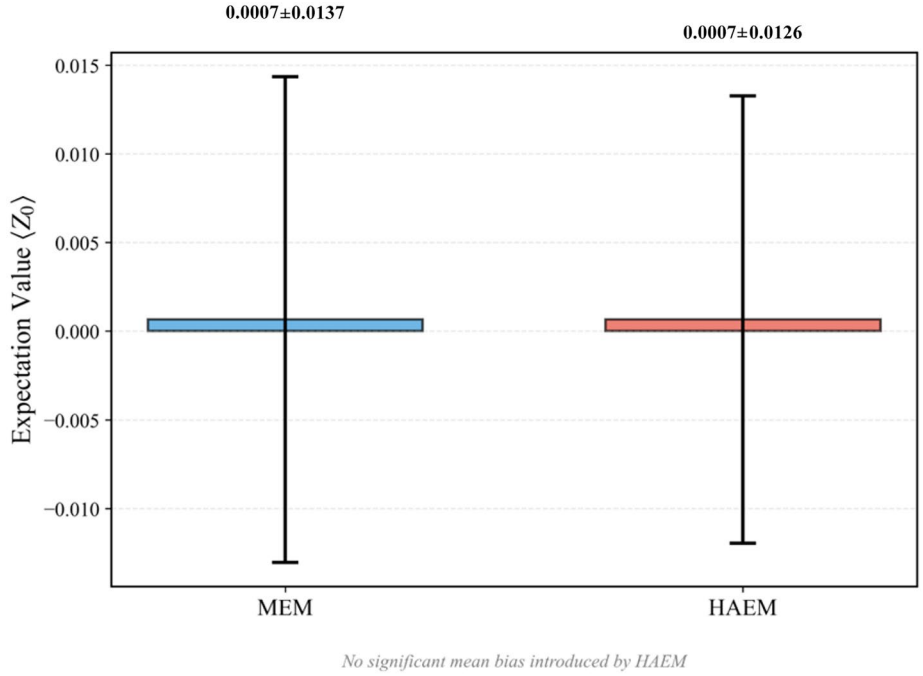


Fig. 2 Shows the average expectation values Z_0 from MEM and HAEM over multiple trials with error bars indicating ± 1 standard deviation

Table 2 Presents the average expectation and standard deviation for the MEM and HAEM

Method	Mean	Standard Deviation	N
Unmitigated	–	–	–
MEM	0.000659	0.013684	20
HAEM	0.000653	0.012607	20

Table 3 Standard deviation and relative variance reduction achieved by HAEM

Method	Standard Deviation	Relative Std Reduction (%)
MEM	0.013684	–
HAEM	0.012607	7.87

4.3 Run-to-run stability analysis

The mean values for both MEM and HAEM are consistent; however, a primary goal of HAEM is to increase run-to-run consistency in the presence of stochastic noise.

Figure 3 depicts the time series of expectation value fluctuations as measured from several independent executions of the two algorithms. The MEM-corrected expectation values show substantial temporal oscillation between successive executions due to the inherent sensitivity of the algorithm to the randomness of the stochastic measurement noise.

HAEM exhibits a modest reduction in oscillatory variation compared to MEM under simulated static noise conditions, along with most of the time series due to the fact that the temporal weighting used within the algorithm utilizes information from the previous executions to lower the degree of abrupt oscillations while still maintaining sufficient responsiveness to the underlying trends of the data.

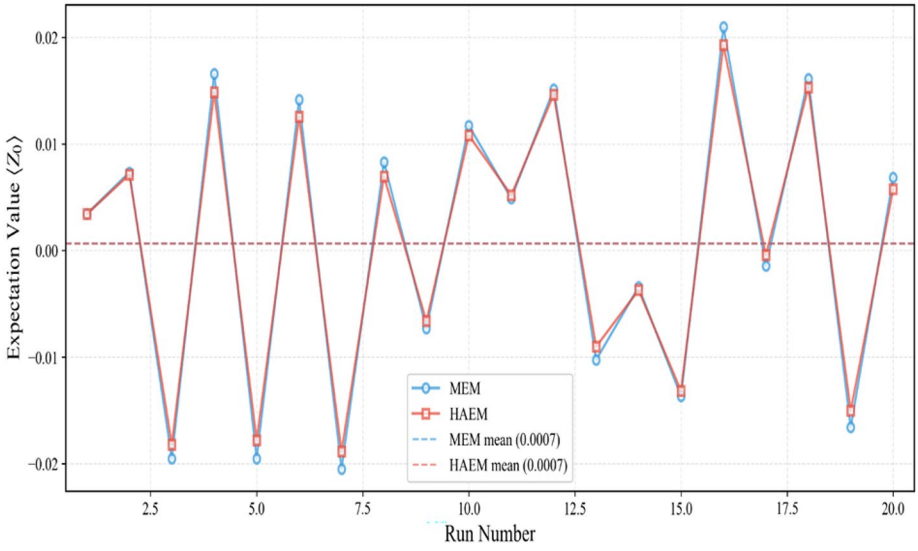


Fig. 3 Run-to-run expectation value variability for MEM and HAEM under repeated simulated executions

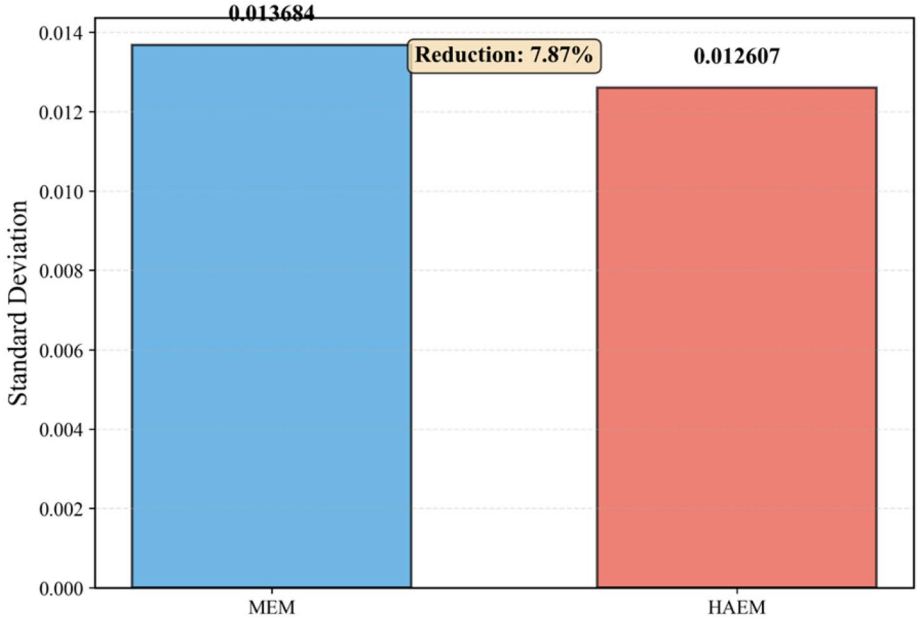


Fig. 4 Comparison of standard deviation for MEM and HAEM

4.4 Features versus noise level

By analyzing the standard deviation of the expectation values calculated for all repeated optimizations, we are able to quantify the improvements in measured stability provided by HAEM versus MEM.

From both Figs. 3 and 4, it is clear that MEM has a Standard Deviation of 0.013684 and HAEM has a Standard Deviation of 0.012607, thus producing a reduction in run-to-run variance of 7.87%. While this relative difference is small in size, it will have a significant impact in the NISQ regime due to the inherent stochastic nature and variability of quantum operations, meaning that many repeated measurements must be performed to produce statistically accurate results.

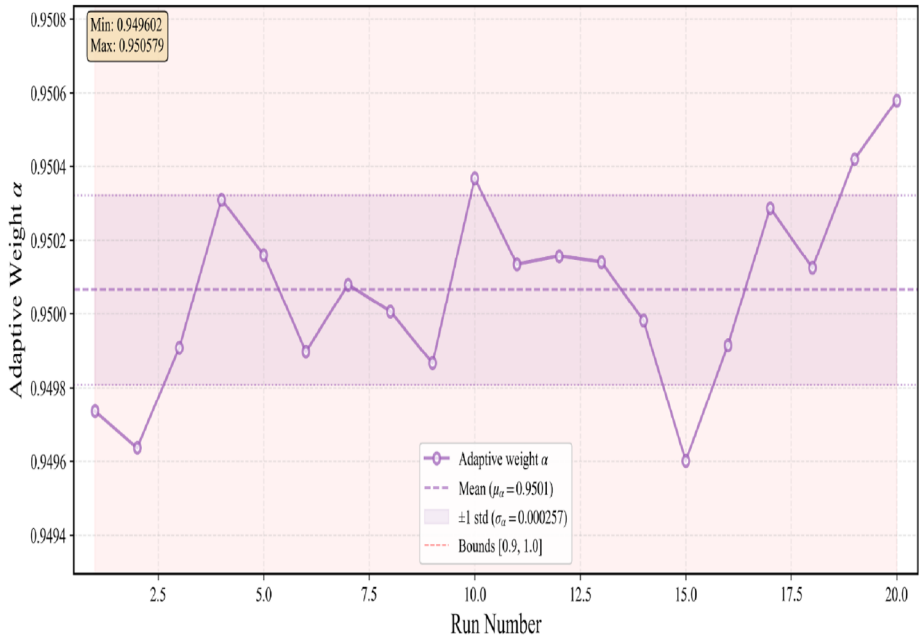


Fig. 5 Shows how much variation exists in value for each execution of the adaptive learning process based on the previous history of adaptive weights

Table 4 Contains the mean, standard deviation, and range of values for adaptive weight α

Mean	Std	Min	Max	Description
0.950065	0.000257	0.949602	0.950579	Adaptive smoothing parameter

Therefore, these results prove the substantial benefits HAEM achieves in producing consistent, observable statistical stability gains beyond the simple elimination of static measurement errors.

4.5 Adaptive weight (α) behavior

To gain insight into the functioning of the adaptive mechanism, the results of the history of the adaptive weight, denoted as α, have been examined over multiple executions.

Figure 5 and Table 4 illustrate a fairly consistent pattern of stability of adaptive weight α over the course of multiple executions. The majority of these values cluster very closely to a point within a very small range of values around α = 0.95 (with an average standard deviation between the iterative runs of on the order of 10–4) resulting in a very narrow range for α. This relatively stable value for α indicates that the adaptive model is performing in a conservative manner thus it does not over-adapt but allows a gradual smoothing effect over time. Further, the consistency of α indicates that the machine-learning component does not add instability and/or oscillation to the performance of the machine; therefore, the performance of HAEM should be regarded as highly reliable when run successively.

Author contributions

D. Lakshanya and P. Nandha Kumar contributed equally to the conception, implementation, experimentation, and manuscript drafting. V. Karthick supervised the project, provided technical guidance, and reviewed the final manuscript.

Data availability

The data supporting the findings of this study are generated through controlled quantum circuit simulations at execution time. The datasets supporting the findings of this work are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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